01 / PHYSICAL RESULT

Clock-Free Physics Cubits

Unknown scales as retained comparisons; a calibrated Planck correspondence

The task is to give every missing quantity its own comparison record, keep numerator and denominator scaling explicit, and combine those records using the native operator algebra. Here a *cubit* means a native two-coordinate comparison cut. It is a mathematical record; interpreting it as a prepared physical qubit requires a separate detector and preparation adapter.

The construction produces a definite physical-scale relation. Let $\ell > 0$ denote an operational length and $E > 0$ an energy assigned to the transport source. Introduce the clock-free combinations

$$A = \hbar c, \qquad B = G/c^4,$$

with units energy×length and length/energy. Keep two positive, dimensionless calibration factors α, β as independent records, and declare the native transport adapter

$$q = \alpha \frac{E\ell}{A}, \qquad p = \beta \frac{BE}{\ell}.$$

The physical interpretation of this adapter is a proposed model choice. The native algebra alone does not select it.

Main conditional result. In the previously derived nontrivial two-cut-compatible cell, positive q, p must equal one. Therefore

$$\boxed{\ell^2 = \frac{\beta}{\alpha}\frac{\hbar G}{c^3}, \qquad E^2 = \frac{1}{\alpha\beta}\frac{\hbar c^5}{G}.}$$

The special normalization $\alpha = \beta = 1$ yields the conventional Planck length and Planck energy. The factors α, β have not been fixed by the existing cell theorem. This is a calibrated correspondence, not a prediction of their values.

What the new construction adds

One unknown	A rank-one comparison projector retaining its signed numerator/denominator ratio.
Separate scaling	A second record compensates independent changes of numerator and denominator scales.
Combined records	Exact tensor composition and product readout, with the discarded coordinates retained as memory.
Physical variables	Four independent unit-invariant combinations for the declared six-scalar dictionary.
Remaining freedom	Exactly two continuous calibration directions remain in (ℓ, E, α, β) at fixed A, B, q, p.
Verification	13 native symbolic certificates, 280 exact checks and 9 negative controls.

Here α denotes a calibration factor, not the electromagnetic fine-structure constant. No numerical value of G, $\hbar$ or c is fitted or predicted in this packet.

02 / ONE UNKNOWN

A native two-record cubit

Keep the canonical native matrices

$$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad K = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Z = -RK = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

They satisfy $R^2 = -\mathbf{1}$, $K^2 = Z^2 = \mathbf{1}$ and $ZK = -KZ$. This representation comes after the inherited cut arithmetic, derived ι, composition and dagger.

Let n, d be real central native coefficients with $d \neq 0$. When they represent physical readings, they must have the same units and be expressed in one common unit chart. Set

$$v(n,d) = \begin{pmatrix} d \\ n \end{pmatrix}, \qquad P(n,d) = \frac{vv^\dagger}{d^2+n^2}, \qquad C(n,d) = 2P(n,d) - \mathbf{1}.$$

S1. Exact comparison cut.

$$C(n,d) = \frac{d^2-n^2}{d^2+n^2}Z + \frac{2dn}{d^2+n^2}K, \qquad C^\dagger = C, \quad C^2 = \mathbf{1}, \quad P^2 = P.$$

Proof. Anticommutation cancels the mixed term in C^2, while $(d^2-n^2)^2 + 4d^2n^2 = (d^2+n^2)^2$. The other identities follow directly. The native compiler verifies the denominator-cleared polynomial identities. □

The ratio is recoverable as $n/d = P_{21}/P_{11}$. Thus an unknown ratio is encoded rather than replaced by a guessed value. Signed and zero numerators are allowed. No probability or Born-rule interpretation is used.

Numerator and denominator need separate scale records

Common rescaling $(n,d) \mapsto (sn, sd)$ leaves P unchanged. Independent rescaling does not: if $n' = an$ and $d' = bd$, then $n'/d' = (a/b)(n/d)$. Retain the scale record $[a : b]$ and compensate it:

$$\boxed{\frac{n}{d} = \frac{n'}{d'}\frac{b}{a}, \qquad P(bn', ad') = P(n,d).}$$

Reciprocal ratios are represented by $KP(n,d)K = P(d,n)$ when $n \neq 0$.

Every reference factor is also visible

For reference readings $\ell_0, E_0, A_0, B_0 > 0$, store

$$r_\ell = \ell/\ell_0, \quad r_E = E/E_0, \quad r_A = A/A_0, \quad r_B = B/B_0.$$

Keep the reference-comparison cubits

$$\eta_Q = \frac{E_0\ell_0}{A_0}, \qquad \eta_G = \frac{B_0E_0}{\ell_0}.$$

The complete factorization, with no reference factor silently set to one, is

$$\epsilon = \eta_Q\frac{r_Er_\ell}{r_A}, \qquad \gamma = \eta_G\frac{r_Br_E}{r_\ell}, \qquad q = \alpha\epsilon, \quad p = \beta\gamma.$$

Each product and reciprocal is implemented by the native record operations. The reference factors and the physical calibration factors have different roles.

03 / COMBINING CUBITS

Products require retained memory

For two records, retain the full four-coordinate tensor

$$v_1 \otimes v_2 = \begin{pmatrix} d_1 d_2 \\ d_1 n_2 \\ n_1 d_2 \\ n_1 n_2 \end{pmatrix}.$$

The product readout and its complementary memory readout are

$$T = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad M = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

Then $T(v_1 \otimes v_2) = (d_1 d_2, n_1 n_2)^T$, representing $(n_1/d_1)(n_2/d_2)$.

S2. Exact product cut and memory completion. With $\mathcal{P} = P_1 \otimes P_2$,

$$\frac{T\mathcal{P}T^\dagger}{\operatorname{tr}(T\mathcal{P}T^\dagger)} = P(n_1 n_2, d_1 d_2), \qquad W = \begin{pmatrix} T \\ M \end{pmatrix}, \quad W^\dagger W = \mathbf{1}_4.$$

The product formula follows by applying T to the rank-one numerator. The second identity follows because W is a permutation. Retaining the whole transformed record preserves all information, including the cross terms between product and memory coordinates. Keeping only two separately normalized diagonal blocks would discard those cross terms.

For the continuation catalogue $K \otimes \mathbf{1}, \mathbf{1} \otimes K$, the row closure of T has rank four. Its initial rank is two; exactly two additional native coordinates are needed for this catalogue. This is a finite memory statement, not a universal bound for arbitrary protocols.

An exact blindness example

The factors $(2, 3)$ and $(1, 6)$ both produce product ratio 6. Their normalized product readouts agree, but their full tensor projectors differ. Hence a combined scalar invariant cannot replace its component records if future operations can inspect those components.

A tensor product alone supplies no interaction

Independent factors obey

$$[R \otimes \mathbf{1}, \mathbf{1} \otimes \iota K] = 0.$$

To use the earlier noncommuting loop theorem, this packet explicitly maps the *derived* ratios q, p into two actions on the same local carrier:

$$U(q) = (\mathbf{1} - qR)^{-1}(\mathbf{1} + qR), \qquad V(p) = (\mathbf{1} - \iota pK)^{-1}(\mathbf{1} + \iota pK).$$

That is the declared coupling law. It does not follow from placing independent comparison records side by side.

The numerical comparison cut $C(q, 1)$ is also distinct from the fixed observer cuts used in the previous sector theorem. Those fixed cuts remain $(\mathbf{1} + K)/2$ and $(\mathbf{1} + \iota R)/2$.

Even central loop signs can be hidden by a composite readout: $(-\mathbf{1}_2) \otimes (-\mathbf{1}_2) = +\mathbf{1}_4$. Factor labels and phase-reference history must be retained when the intended target distinguishes the individual loops. Tensor composition here is distinct from graph gluing in the previous note.

04 / PHYSICAL DICTIONARY

Four invariant coordinates, with no clock

The native construction accepts abstract unit charges. For a concrete physics adapter, choose length and energy as independent unit labels. The declared scalar dictionary is

Quantity	Meaning	Length charge	Energy charge
ℓ	Operational length	1	0
E	Transport-source energy	0	1
$A = \hbar c$	Quantum conversion combination	1	1
$B = G/c^4$	Gravitational conversion combination	1	-1
$\mathscr{R}$	Proposed scalar curvature reading	-2	0
Λ	Proposed cosmological curvature scale	-2	0

S3. Complete monomial unit invariants for this dictionary. The unit-charge matrix has rank two:

$$D = \begin{pmatrix} 1 & 0 & 1 & 1 & -2 & -2 \\ 0 & 1 & 1 & -1 & 0 & 0 \end{pmatrix}.$$

Its integer kernel is generated by the exponent vectors of

$$\boxed{\epsilon = \frac{E\ell}{A}, \quad \gamma = \frac{BE}{\ell}, \quad \kappa = \mathscr{R}\ell^2, \quad z_\Lambda = \Lambda\ell^2.}$$

Proof. For an exponent vector $(u_\ell, u_E, u_A, u_B, u_R, u_\Lambda)$, the two zero-charge equations give $u_E = -u_A + u_B$ and $u_\ell = -u_A - u_B + 2u_R + 2u_\Lambda$. The corresponding monomial is $\epsilon^{-u_A}\gamma^{u_B}\kappa^{u_R}z_\Lambda^{u_\Lambda}$ on its nonzero domain. This also proves independence. Native exact row reduction verifies rank and the displayed kernel. □

This is completeness for the chosen six *scalar* quantities. A metric, connection, tensor source or field of values requires its own components and compatibility laws; four numbers do not constitute a complete gravitational theory.

Energy cancels in one comparison; length in the other

$$\frac{\gamma}{\epsilon} = \frac{AB}{\ell^2}, \qquad \epsilon\gamma = \frac{BE^2}{A}.$$

The conventional physical adapter identifies

$$AB = \frac{\hbar G}{c^3} = \ell_P^2, \qquad \frac{A}{B} = \frac{\hbar c^5}{G} = E_P^2.$$

These conventional Planck definitions are documented by CODATA/NIST [1]. Their appearance here is algebraic correspondence under the dictionary, not a new measurement of fundamental constants.

The unresolved clock coordinate is explicit

For any $\tau > 0$, the replacement

$$c \mapsto \tau c, \qquad \hbar \mapsto \hbar/\tau, \qquad G \mapsto \tau^4 G$$

preserves A and B. Thus this clock-free dictionary cannot separately identify $c, \hbar, G$. No physical time variable is needed for the retained comparisons; separating those constants requires an additional adapter that distinguishes this family.

05 / SECTOR TO PHYSICS

The calibrated scale theorem

The previous cell uses two independent central Cayley parameters q, p and the loop

$$H = V(p)^\dagger U(q)^\dagger V(p) U(q).$$

Its invariant defect is

$$\chi = \frac{16 q^2 p^2}{(1+q^2)^2(1+p^2)^2}, \qquad (H-\mathbf{1})^\dagger (H-\mathbf{1}) = 4\chi \mathbf{1}.$$

Compatibility with both fixed noncommuting observer cuts forces $H = \pm\mathbf{1}$. In the positive nontrivial sector, $q = p = 1$ and $\chi = 1$. This premise is inherited with its original four-cycle, local-rank and observation conditions.

S4. Physical-scale relation under the declared adapter. With $q = \alpha E \ell / A$, $p = \beta B E / \ell$ and all six quantities positive, the selected sector implies

$$\boxed{A = \alpha E \ell, \qquad B = \frac{\ell}{\beta E}.}$$

Consequently,

$$\boxed{\alpha \ell^2 = \beta A B, \qquad \alpha\beta B E^2 = A.}$$

The positive length and energy are therefore

$$\ell = \sqrt{\frac{\beta}{\alpha}}\, \ell_P, \qquad E = \frac{E_P}{\sqrt{\alpha\beta}}.$$

Square roots use the already admitted native real completion. The executable verifies the rational polynomial relations exactly.

A symbolic proof that retains the residuals

Define the unclosed comparison residuals

$$F_Q = \alpha E \ell - A, \qquad F_G = \beta B E - \ell.$$

Before assuming either residual vanishes, native polynomial reduction proves

$$\alpha \ell^2 - \beta A B = \beta B F_Q - \alpha \ell F_G,$$

$$\alpha\beta B E^2 - A = F_Q + \alpha E F_G.$$

These identities provide the exact implication when $F_Q = F_G = 0$ and preserve the defects when the two comparisons are not closed. They are more informative than setting every unknown scale to one.

What can be solved in physical notation?

If the required readings and calibration factors are independently supplied, the same relation gives

$$\hbar = \frac{\alpha E \ell}{c}, \qquad G = \frac{c^4 \ell}{\beta E}.$$

These are conditional reconstruction formulas. The conventional labels G and $\hbar$ enter through the declared physics dictionary; no gravitational field equation has been deduced from a relabelling of the native loop.

For $\alpha = \beta = 1$, the selected branch has $(\ell, E) = (\ell_P, E_P)$. This is a useful special case to investigate. Selecting that calibration or proving that a physical probe implements it remains an additional research step.

06 / WHAT IS STILL UNKNOWN

The remaining freedom has rank two

S5. Exact calibration-family theorem. At fixed A, B, the native parameters q, p are unchanged under

$$\boxed{\ell \mapsto s\ell, \quad E \mapsto tE, \quad \alpha \mapsto \frac{\alpha}{st}, \quad \beta \mapsto \frac{s\beta}{t}, \qquad s, t > 0.}$$

This is not merely a change of physical units: A, B are held fixed, while the assigned physical readings and adapter factors vary together.

Proof. Substitution cancels s, t in both $\alpha E\ell/A$ and $\beta BE/\ell$. Conversely, for any positive ℓ, E and fixed q, p, A, B,

$$\alpha = \frac{qA}{E\ell}, \qquad \beta = \frac{p\ell}{BE}.$$

Thus ℓ, E parameterize the whole positive solution family. In logarithmic coordinates $(\log \ell, \log E, \log \alpha, \log \beta)$, the observation matrix is

$$J = \begin{pmatrix} 1 & 1 & 1 & 0 \\ -1 & 1 & 0 & 1 \end{pmatrix}, \qquad \operatorname{rank} J = 2,$$

with kernel basis $(1, 0, -1, 1)$ and $(0, 1, -1, -1)$. □

A concrete exact counterexample

In one common length/energy unit chart set $A = B = 1$:

Case	ℓ	E	α	β	(q, p)	H
I	1	1	1	1	$(1, 1)$	$\mathbf{-1}$
II	2	3	1/6	2/3	$(1, 1)$	$\mathbf{-1}$

Both cases give the same native cell operator, loop defect and cost levels. The calibration cubits differ and therefore expose the ambiguity. Erasing them would make the physical scale appear determined when it is not.

The executable verifies nine such selected-sector examples. The general family above, rather than the finite sample, proves the two remaining degrees of freedom.

What new information would close this family?

Two independent constraints transverse to this family are sufficient within this scalar adapter. For example, separately established α, β determine ℓ, E; separately calibrated ℓ, E determine α, β. One relation fixing β/α fixes the length ratio ℓ/ℓ_P; a second fixing $\alpha\beta$ fixes E/E_P.

The appropriate target is therefore specific: derive or measure the *relative calibration* and the *product calibration*. Merely adding more encodings of the same two observations cannot increase their rank.

Encoding an unknown preserves it for later inference. A new value is selected only when a new independent relation, observation or constitutive condition acts on that record. The present result locates those missing relations exactly.

07 / CURVATURE AND ENERGY

Additional physical ports stay explicit

A finite loop-defect curvature proxy

Let the face area be $a_f = \omega\ell^2$, with a positive shape factor ω retained as another comparison cubit. Define the finite positive defect magnitude by

$$\mathcal{K}_\Delta^2 = \frac{1}{8a_f^2}\,\mathrm{tr}\big[(H-\mathbf{1})^\dagger(H-\mathbf{1})\big] = \frac{\chi}{a_f^2}.$$

This is a definition of a finite-loop observable with curvature-squared units. It is not automatically Ricci scalar curvature or an infinitesimal connection curvature.

S6. Selected-sector defect/scale relation. At $\chi = 1$, the positive root gives

$$\mathcal{K}_\Delta \ell^2 = \frac{1}{\omega}, \qquad \boxed{\mathcal{K}_\Delta \ell_P^2 = \frac{\alpha}{\beta\omega}.}$$

This follows from the selected length relation. The executable checks its squared rational form for all nine calibration examples.

A further proposed identification $\mathscr{R} = \zeta\mathcal{K}_\Delta$ would introduce an independent signed calibration record ζ. It would give $\mathscr{R}\ell^2 = \zeta/\omega$, but would still require a tensor/coframe and physical-probe justification. The sign and tensor content of gravitational curvature cannot be reconstructed from a positive defect norm alone.

The cosmological record has not acquired a field equation

One can store $z_\Lambda = \Lambda\ell^2$ using the same cubit. The current loop law contains no Λ term. Changing this record therefore leaves H unchanged; there is no prediction of Λ until a justified coupling to the retained geometry is added.

Transport-source energy and operator levels are different roles

The source scale E in q, p is not an eigenvalue of the cell cost operator. The inherited operator still has

$$\mathcal{L}^2 - 4\mathcal{L} + 2\mathbf{1} = 0, \qquad \lambda_\pm = 2 \pm \sqrt{2}.$$

An affine physical energy interpretation would need its own origin and gain:

$$\mathsf{H}_{\mathrm{phys}} = E_{\mathrm{off}}\mathbf{1} + \kappa_E\mathcal{L}, \qquad \Delta E = 2\sqrt{2}\,\kappa_E.$$

Neither E_{off} nor κ_E is fixed by encoding E. They can be retained as comparison records, and the gap needs only the gain. The ratio $3 + 2\sqrt{2}$ transfers to physical energies only with a specified zero origin and proportional energy adapter.

The next physics-facing closure

The remaining task is to obtain an independently justified relation for β/α and $\alpha\beta$, together with the source-response and curvature adapters. The native boundary-response/memory work supplies a place to test such relations without discarding invisible modes. A larger tensor of unknowns alone does not supply the missing interaction law.

08 / CERTIFICATES AND SOURCES

Reproducible scope

Exact checks completed

13 symbolic certificates	Comparison circle, self-dagger cut, involution, projector, common scaling, reciprocal, tensor relations, independent scale correction and physical residual identities. Each certificate is serialized and replayed.
280 exact checks	Includes 18 signed/zero comparison records; tensor product readouts; memory closure; reference-factor composition; invariant ranks; unit rescaling; calibration families; and the inherited cell polynomial.
9 negative controls	Product blindness; uncancelled independent scaling; commuting independent factors; tensor sign cancellation; unfixed physical calibrations; their distinction by retained cubits; clock ambiguity; uncoupled Λ; and source energy versus cost levels.
Native implementation	Unchanged RKF native arithmetic and Paninian compiler. Python 3.12 performs only integrity checks and process orchestration.

The symbolic proofs establish equality in their declared native quotients. Written arguments establish the completeness and calibration-family claims. These checks are not external peer review, experimental quantum-gravity validation or a claim of Lean formalization.

Lineage and physical attribution

The complete preceding *Clock-Free Cut Quantisation* packet is included under `parent/`, with its original sources and hashes. Its inherited baseline includes the native algebra, the gravity-before-curvature type bridge, source-response laws, and the phase-to-response translator. Those existing results already distinguish a coherent algebraic law from physical source and scale calibration. This note extends that distinction into explicit comparison records and a solvable calibration family.

Canonical native source: Recognition-Kernel-Framework main, `e8e3089745bf7dc0532d01c62a5f660f5658c6f0`. No repository code is altered. The imported scalar engine is not duplicated by an alternative mathematical backend.

[1] **Conventional physical adapter.** P. J. Mohr et al., *CODATA recommended values of the fundamental physical constants: 2022*, Reviews of Modern Physics 97, 025002 (2025), Table XXXIII, printed page 45. NIST-hosted source: physics.nist.gov/cuu/pdf/RevModPhys.97.025002.pdf. Used for the conventional dimensions and Planck combinations; no numerical fit or measured constant is imported into the native certificates. The algebraic appearance of these known combinations is not a claim to have discovered their definitions.

Run the packet

Unzip the source archive and run `python3.12 verify.py`. It verifies the current and inherited manifests and launches `node verify_scale_cubits.cjs -check`. Replay is read-only and compares the entire new result with `SCALE_CUBIT_RESULT.json`.

`PHYSICS_ADAPTERS.json` records the conventions, sources and unclosed physical ports. `README.txt` explains reproduction. The editable LaTeX note, native verifier, exact records, certificate objects, prior packet and SHA-256 manifest are all included.

Established result: a native calculus for retaining unknown scales, composing their invariants with memory, and deriving the physical-scale relations of the selected finite cell under an explicit adapter. Physical quantum gravity requires the remaining calibration, interaction and geometry laws.